

HTE reading group

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Paper

Generic machine learning inference on heterogeneous treatment effects in randomized experiments, with an application to immunization in India

by Victor Chernozhukov, Mert Demirer, Esther Duflo, and Iván Fernández-Val

775 citations on Google Scholar.

Focus

Conceptual focus.

Less focus on the actual estimators and the technical requirements for their validity.

More focus on the interpretation of the estimand and null hypotheses that these estimators targets.

Major points

Target **features** of CATE

Estimands that depend on machine learning **proxies** of CATE

Insights on best **approximations** of CATE (in **RCTs!**)

Motivation

Investigate/test for heterogeneity of treatment in RCT setting with high-dimensional covariate vector. Estimation of CATE (or even low-dimensional features of it) is too hard.

Without some form of sparsity, it is hard, if not impossible, to obtain consistent estimators of [CATE]. . . . One fundamental reason is that ML methods might not even produce consistent estimators of [CATE] in high dimensional settings. . . . Hence, generic ML estimators cannot be regarded as consistent, unless further assumptions are made. Examples of such assumptions include structured forms of linear and non-linear sparsity and super-smoothness. The problem of obtaining uniformly valid inference on [CATE] using generic ML methods is even more difficult.

In this paper, we take an agnostic view. We neither rely on any sparsity or low-complexity assumptions to make the ML estimators consistent, nor impose other stronger conditions to make “traditional” confidence intervals valid. We simply treat ML as providing proxy predictors for the objects of interest. (p.8)

Some notation and setup

Potential outcomes $Y(0)$ and $Y(1)$, binary treatment indicator D , and high-dimensional covariate vector Z .

Main causal functions (identified through standard assumptions) are

$$b_0(z) = b_P(z) = \mathbb{E}[Y(0) \mid Z = z] = \mathbb{E}[Y \mid D = 0, Z = z]$$

and the CATE function

$$s_0(z) = s_P(z) = \mathbb{E}[Y(1) - Y(0) \mid Z = z] = \mathbb{E}[Y(0) \mid Z = z] - \mathbb{E}[Y \mid D = 0, Z = z]$$

The propensity score

$$p(z) = P(D = 1 \mid Z) = P(D = 1 \mid Z_1)$$

is **known**, i.e., RCT setting.

$B(z) = B_a(z) = B(z; \text{Data})$ and $S(z) = S_a(z) = S(z; \text{Data})$ denote **machine learning proxies** for b_P and s_P .

Stage 1 (p.9)

Let (M, A) denote a random partition of the set of indices $\{1, \dots, N\}$. The strategies that we consider rely on random splitting of $\text{Data} = (Y_i, D_i, Z_i)_{i=1}^N$ into a main sample, denoted by $\text{Data}_M = (Y_i, D_i, Z_i)_{i \in M}$, and an auxiliary sample, denoted by $\text{Data}_A = (Y_i, D_i, Z_i)_{i \in A}$. We will sometimes refer to these samples as M and A . After splitting the sample, we carry out two stages:

Stage 1: From the auxiliary sample A , we obtain ML estimators of the baseline functions and treatment effects, which we call the ML proxy predictors,

$$z \mapsto B(z) \text{ and } z \mapsto S(z).$$

Here $S(Z)$ is a possibly biased and noisy predictor of $s_0(z)$ and $B(Z)$ is a possibly biased and noisy predictor of $b_0(Z)$ (or other technical “baseline” functions, as we discuss in Section 5). We do not require these predictors to be consistent for the true functions.

Stage 2 (p.9)

Stage 2: We post-process the proxies from Stage 1 to estimate and make inference on features of the CATE function $z \mapsto s_0(z)$ in the main sample M . The key features that we target include:

- (1) **Best Linear Predictor** (BLP) of the CATE $s_0(Z)$ on the ML proxy predictor $S(Z)$;
- (2) **Sorted Group Average Treatment Effects** (GATES): average of $s_0(Z)$ (ATE) by heterogeneity groups induced by the ML proxy predictor $S(Z)$;
- (3) **Classification Analysis** (CLAN): average characteristics of the most and least affected units defined in terms of the ML proxy predictor $S(Z)$.

Our approach is *generic* with respect to the ML method being used, and is *agnostic* about its formal properties. However, it relies on sample splitting, which introduces a specific source of uncertainty. To account for this, we use many data splits into main and auxiliary samples to produce robust estimators, and we employ quantile aggregation of inference to combine results across splits.

Summary of the proposal

Split data into main and auxiliary samples.

Stage 1 Fit complex ML algorithms for estimating b_0 and s_0 in the auxiliary sample, obtaining the proxies B and S .

Stage 2 Consider B and S as fixed functions and estimate some low-dimensional feature of them and the true CATE s_0 in the main sample.

Repeat and base inference on quantiles obtained across different splits into main and auxiliary samples.

Conditional (data-adaptive) estimands (p.12)

In this section we condition on Data_A (the auxiliary sample in a random split) and therefore consider the functions

$$z \mapsto B_A(z) := B(z; \text{Data}_A) \text{ and } z \mapsto S_A(z) := S(z; \text{Data}_A).$$

as fixed functions. To lighten the notation, we keep the dependence on Data_A implicit in all the expectations, objects and parameters, that is, e.g., we use $E[\cdot]$ and $S(Z)$ instead of $E[\cdot \mid \text{Data}_A]$ and $S_A(Z)$. We shall make the dependence on Data_A explicit when we discuss estimation and inference in Section [4](#).

Example of feature: Best Linear Prediction (BLP) (compare p.12)

Definition (BLP)

The best linear predictor of $s_P(Z)$ by $S(Z) = s_A(Z) = S(Z; \text{Data}_A)$ is the solution to

$$\min_{b_1 \in \mathbb{R}, b_2 \in \mathbb{R}} \mathbb{E}_P[(s_P(Z) - b_1 - b_2 S(Z; \text{Data}_A))^2 \mid \text{Data}_A],$$

which, if exists, is defined as

$$\text{BLP}_P[s_P(Z) \mid S(Z; \text{Data}_A)] := \beta_1 + \beta_2(S(Z; \text{Data}_A) - \mathbb{E}_P[S(Z; \text{Data}_A) \mid \text{Data}_A]),$$

where $\beta_1 = \mathbb{E}_P[s_P(Z)]$ and

$$\beta_2 = \text{Cov}_P[s_P(Z), S(Z; \text{Data}_A) \mid \text{Data}_A] / \text{Var}_P[S(Z; \text{Data}_A) \mid \text{Data}_A].$$

I write $s_P = s_0$ and $\mathbb{E}_P = \mathbb{E}$ to make it clear that the “true” expectation $\mathbb{E}$ means the expectation under P according to which the “true” CATE s_0 is defined.

Why conditioning on data?

They are discussing **two** things to address the “*fundamental impossibilities in non-parametric inference [of CATE estimation]*” (p.3).

Firstly: Target low-dimensional features of CATE, e.g., BLP, GATE, CLAN.

Why not stop here?

- Would require that the CATE function can be approximated fast enough, typically at a rate which should be $o_P(n^{-1/4})$.
- This is unrealistic in truly high-dimensional settings (i.e., realistic finite-sample scenario).

Secondly: Target low-dimensional features of **data-dependent proxies** of CATE and make inference **conditional on data**.

Trading unachievable rate-convergence criteria for post-selection inference interpretations.

Definition and interpretation of the estimand (p.21)

Definition 4.1 (Collection of Splits). *Consider the collection $\{(a, m), a \in \mathcal{A}\}$ of partitions of $[N] = \{1, \dots, N\}$ into auxiliary sets a of cardinality $N - n$ and main sets m of cardinality n . We generate the collection independently of*

$$\text{Data} := (Y_i, D_i, X_i)_{i=1}^N.$$

Different partitions of $[N]$ yield different estimands and estimators. To formalize this randomness, we consider A as a uniform random variable taking values $a \in \mathcal{A}$, that is, $A \sim U(\mathcal{A})$. Therefore, conditional on Data, the estimand θ_A is a random variable. In what follows, we will mainly target our inference on the median value of θ_A , instead of θ_a , the estimand for a specific partition.

Definition and interpretation of the estimand – my attempt

Define the data-dependent parameter $\theta(P, \text{Data}_a)$ as

$$\theta(P; S, \text{Data}_a) = \text{BLP}_P[s_0(Z) \mid S(Z; \text{Data}_a)].$$

The data-dependent “median” parameter $\theta_M(P, \text{Data})$ is defined as

$$\theta_M(P; S, \text{Data}) = \text{Median}(\theta(P; S, \text{Data}_A) \mid \text{Data}) = \text{Median}_{A \sim U(\mathcal{A})}(\theta(P; S, \text{Data}_A) \mid \text{Data}).$$

Would the parameter $\theta(P; S, \text{Data})$ be more interesting? Or $\mathbb{E}[\theta(P; S, \text{Data})]$?

Would we expect the parameter $\theta_M(P; \text{Data})$ to be close to $\theta(P; \text{Data})$ or $\mathbb{E}[\theta(P; S, \text{Data})]$?

Can we relate the interpretation of this to the type of performance measures we get from cross-validation?

Interpretation of null hypothesis

4.2. Hypothesis Testing with Multiple Splits. We start the analysis by testing homogeneous hypotheses $\theta_A = \theta_0$, which imply that θ_a does not vary with a . Suppose, for example, that we want to test that the slope of the BLP is zero with probability one, $\beta_{2A} = 0$, against the alternative $\beta_{2A} > 0$ with positive probability. This problem amounts to both testing the heterogeneity in CATE and the relevance of the ML score S_A as a predictor. Another interesting hypothesis is whether $\beta_{2A} = 1$, with probability one, that is, whether S_A is well-calibrated and needs no post-processing.

More generally, suppose we are testing the hypothesis

$$H_0 : \theta_A = \theta_0, \tag{4.6}$$

with probability one, against $H_1^+ : \theta_A > \theta_0$ with positive probability. Testing using the median

Applied analysis

Immunization (vaccination?) of children i North India (against measles?).

Interpretation of results

TABLE 2. Comparison of Causal ML Methods: Immunization Incentives

	Elastic Net	Boosting	Neural Network	Random Forest
Best BLP (Λ)	67.750 [51.491, 82.368]	32.900 [23.246, 44.665]	53.420 [42.516, 67.647]	25.200 [18.328, 34.705]
Best GATES ($\bar{\Lambda}$)	8.254 [7.329, 9.314]	5.104 [4.27, 6.079]	6.001 [5.087, 6.888]	4.492 [3.339, 5.507]

Notes: Medians over 250 splits. Note that we used Neural Network Causal Boosting for all methods, using Algorithm [6.2](#) The brackets report interquartile ranges for goodness-of-fit statistics.

Interpretation of results

TABLE 3. BLP of Immunization Incentives Using Causal Proxies

Elastic Net		Neural Network	
ATE (β_1)	HET (β_2)	ATE (β_1)	HET (β_2)
2.814	1.047	2.441	0.899
(1.087,4.506)	(0.826,1.262)	(0.846,3.979)	(0.685,1.107)
[0.004]	[0.000]	[0.004]	[0.000]

Notes: Medians over 250 splits. Median Confidence Intervals ($\alpha = .05$) in parenthesis. P-values for the hypothesis that the parameter is equal to zero against the two-sided alternative in brackets.

Table 3 presents the results of the BLP of CATE on the ML proxies. We report estimates of the coefficients β_1 and β_2 , which correspond to the ATE and heterogeneity loading (HET) parameters in the BLP. The ATE estimates in Columns 1 and 3 indicate that the package treatment increases the number of immunized children by 2.81 based on elastic net estimates and by 2.44 based on neural network estimates. Reassuringly, these estimates are on either side of the raw difference in means (2.77). Focusing on the HET estimates, we find strong heterogeneity in treatment effects, as indicated by the statistically significant estimates. Moreover, the estimates are close to 1, suggesting that the ML proxies are good predictors of the CATE.

Addendum: Best approximation of CATE (in RCTs)

Getting the best approximation of CATE s_0 based on the proxy S is not just OLS – need to use weight based on the propensity score.

Alternatives for building causal learners of CATE.

These tricks only seem to work in an RCT setting(?). So comparison with existing method, which may be developed for observational settings, are perhaps not fair.